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Modality Alignment-Driven large language model for wind farm power forecasting

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Abstract

Accurate forecasting of wind farm power output plays a vital role in grid dispatching, optimizing transactions in electricity markets, and enhancing the overall operational security of power systems. In recent years, large language models (LLMs) have been introduced into wind power forecasting due to their strong reasoning capabilities and rich domain knowledge. However, applying LLMs in this domain remains challenging, primarily because they require substantial amounts of data and impose stringent deployment requirements. In practice, the harsh environments in which wind farms operate make it difficult to collect sufficient power data to meet the demands of LLM training. Moreover, real-world deployment often falls short of the computational and infrastructural requirements these models entail. As a result, LLM-based forecasting methods have struggled to meet the expectations for accuracy and practicality in real scenarios. To address these challenges, a novel modality alignment-driven large language model (MALLM) for wind farm power forecasting is proposed. The proposed model leverages a modality alignment method to resolve the mismatch between the modality of power data and the input expectations of LLMs. It eliminates the need for retraining or fine-tuning the language model, significantly reducing both deployment costs and data requirements. Furthermore, the modality alignment method incorporates temporal information embedding, enabling the integration of aligned time-related features with power data. This helps preserve key temporal patterns that are often lost during standard modality transformation processes. Comparative and ablation experiments conducted on supervisory control and data acquisition (SCADA) datasets from two wind farms located in distinct geographical regions demonstrate the effectiveness and robustness of the proposed method.

Keywords: Wind power forecasting; Deep learning; Large language model; Modality alignment; Time information.

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1 Introduction

With the global energy transition, wind power, as a clean and renewable energy source, has been playing an increasingly important role in power systems [1-3]. However, the intermittency and randomness of wind result in significant fluctuations and uncertainties in wind power generation. This intrinsic unpredictability poses significant challenges to the stable operation of power systems after wind power integration [4]. To ensure secure and stable grid operation, enhance scheduling optimization, and improve renewable energy integration, power systems are increasingly dependent on accurate wind power forecasting. High-precision wind power forecasting facilitates rational allocation of reserve capacity, optimizes unit scheduling, and reduces reserve costs and wind curtailment caused by forecasting errors, thereby enhancing both the reliability and economic efficiency of wind power integration [5].

Nevertheless, achieving accurate wind power forecasting remains challenging, primarily due to the stochastic and non-stationary nature of wind power output and the interference of data noise [6]. On the one hand, wind power time series exhibit non-stationarity: their statistical characteristics vary with weather, seasons, and geographical conditions, making it difficult for traditional forecasting models to adapt to such dynamic variations [7, 8]. On the other hand, SCADA monitoring data from wind farms frequently contain outliers, missing values, and measurement errors. These data characteristics increase modeling complexity, necessitating forecasting methods that can both suppress noise interference and capture underlying complex patterns. The strong stochastic fluctuations of wind power, combined with data quality issues, render the development of robust and accurate forecasting models a highly challenging task.

In recent years, researchers have proposed various approaches for wind power forecasting. The commonly used methods can be categorized into statistical models [9], physical models [10], and artificial intelligence (AI)-based methods [11]. Traditional statistical models, such as

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Autoregressive Moving Average (ARMA) [12], Autoregressive Integrated Moving Average (ARIMA) [13], Vector Autoregression (VAR) [14], rely on assumed linear relationships to fit historical data. However, statistical models fail to effectively capture the nonlinear and non-stationary characteristics of wind power series, resulting in forecasting accuracy that often falls short of practical operational requirements. Physical models, such as numerical weather prediction (NWP), forecast wind power by simulating atmospheric physical processes. However, their limited spatial resolution makes it difficult to capture rapid local variations in wind power [15]. Machine learning methods, such as support vector machines and random forests, can capture nonlinear relationships to some extent and offer better short-term forecasting performance than statistical models. However, they rely on handcrafted feature extraction and exhibit limited generalization capability [16]. In recent years, deep learning models have achieved remarkable progress in wind power forecasting. For example, Refs. [17, 18], applied convolutional neural networks (CNNs) to wind power forecasting and achieved promising results. However, CNNs rely on receptive fields of convolutional kernels to extract local features, which makes them effective at capturing short-term patterns but insufficient for modeling the long-term dependencies inherent in wind power series. To address this issue, researchers have proposed new neural architectures such as Long Short-Term Memory (LSTM) networks [19] and Gated Recurrent Units (GRUs) [20]. López et al. [21] validated the superiority of LSTM in wind speed forecasting, demonstrating that its forecasting errors were significantly lower than those of CNNs and RNNs. Compared with CNNs and recurrent neural networks (RNNs), LSTM dynamically regulates the retention and discarding of historical information through forget and update gates. This enables LSTM to preserve and update information over longer time horizons, thereby capturing deeper data features in highly stochastic and volatile wind power series. Although LSTM and GRU can handle temporal dependencies, they tend to forget earlier information when processing long wind power sequences, thereby limiting their capability for long-term modeling. The Transformer model [22], based on the self-attention mechanism, can simultaneously capture global dependencies and local dynamics. It not only enhances the modeling of long-term trends and sudden fluctuations but also offers efficient parallelization and cross-site generalization advantages, making it more suitable than CNN, LSTM, and GRU for large-scale, multi-scenario wind power forecasting tasks [23].

Today, the rise of large language models (LLMs) has created new opportunities for time series forecasting. Benefiting from pretraining on massive text corpora, LLMs possess powerful reasoning capabilities and extensive knowledge reserves [24]. Unlike traditional deep learning models based on autoregressive principles, LLMs can exploit their intrinsic reasoning ability to infer deep features from power data, thereby achieving more accurate forecasting results [25]. LLM-based wind power forecasting methods provide a new pathway for improving forecasting accuracy. Methods for wind power forecasting based on large language models (LLMs) can be broadly categorized into two types: LLM retraining or fine-tuning approaches and data modality adaptive approaches. Both types effectively leverage the powerful reasoning capability of LLMs to achieve accurate wind power forecasting. Since the text corpora used to train LLMs are in a discrete modality while wind power data are continuous, directly feeding such data into LLMs for forecasting can lead to erroneous reasoning results. To address this, Lai et al. [26] fine-tuned the BERT model and embedded spatiotemporal relationships of wind speed data into the forecasting process, achieving higher accuracy than commonly used time series models. Wu et al. [27], in addition to fine-tuning LLMs, introduced a spatiotemporal prompting module to strengthen their ability to reason about both global and local features of wind speed data. Although fine-tuning LLMs can mitigate the accuracy degradation caused by modality mismatches, their massive parameter scales mean that the data volume and computational resources required for fine-tuning are often unattainable in harsh wind farm operating environments. Beyond fine-tuning, another approach is to align the modality of wind power data with the modality that LLMs can recognize, thereby enabling direct forecasting using pretrained LLMs. This strategy eliminates the need for retraining or fine-tuning and addresses forecasting errors arising from modality mismatches from the perspective of data processing [28].

To this end, this paper proposes a modality alignment-driven large language model (MALLM) for wind farm power forecasting, addressing the challenges of insufficient data and high deployment costs. The modality alignment module matches wind power data with the modality required by LLMs, thereby avoiding the need for retraining or fine-tuning. Furthermore, temporal information is embedded into the aligned power data to mitigate information loss during modality alignment, thereby enhancing the ability of LLMs to capture periodic patterns in SCADA data. Comparative and ablation experiments conducted on SCADA datasets from two geographically distinct wind farms confirm the effectiveness of the proposed method.

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The main contributions of this paper are summarized as follows:

(1) A novel modality alignment-driven large language model (MALLM) for wind farm power forecasting is proposed. Through temporal information integration during modality alignment, the method fully exploits the reasoning capability of LLMs to capture periodic patterns in SCADA data without retraining or fine-tuning.

(2) A wind power data modality alignment method based on temporal information embedding is proposed. This approach aligns wind power data with the discrete textual modality recognizable by LLMs while embedding temporal information into the transformed data, enabling LLMs to accurately interpret the input and enhancing their ability to capture temporal features.

(3) To evaluate the forecasting performance of MALLM, comprehensive comparative and ablation experiments were conducted on SCADA datasets from two real-world wind farms in geographically distinct regions. Experimental results demonstrate that the proposed method consistently outperforms state-of-the-art time series models across various forecasting horizons on both datasets.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 provides a detailed description of the proposed model and its methodology; Section 4 validates the effectiveness of the model using SCADA data from two wind farms located in different regions; and Section 5 concludes the study.

2 Related Work

2.1 Wind power forecasting

Over the past decades, wind power forecasting technology has made substantial progress. With the continuous advancement of computer science, computational capacity has significantly improved, and deep learning models exhibit strong feature extraction capability for non-stationary wind power data. Consequently, AI-based approaches have been widely applied in wind power forecasting [29]. In the early stages of AI development, simple neural networks such as CNNs and RNNs were predominantly employed. Zhao et al. [30] applied CNNs to short-term wind power forecasting and achieved favorable results. Cai et al. [31] employed RNNs for wind speed forecasting, and the results indicated relatively high forecasting accuracy. Li et al. [32] applied LSTM to offshore wind speed forecasting, and the results demonstrated that LSTM achieved higher forecasting accuracy than CNNs and RNNs. Because wind speed and power data are highly stochastic and non-stationary, simply increasing the depth of neural networks is insufficient to substantially improve forecasting accuracy. The Transformer model introduces a self-attention mechanism, enabling simultaneous extraction of local and global features from long sequences while avoiding issues such as gradient vanishing and training inefficiency that may arise in deep neural networks [33].

Li et al. [34] treated wind power data as a signal processing task and combined the empirical mode decomposition (EMD) algorithm with the Transformer model, showing that EMD-preprocessed data yielded higher forecasting accuracy. Michalakopoulos et al. [35] developed a physics-informed Transformer model that integrates geographic and spatial information into the forecasting process, thereby improving feature extraction accuracy. To effectively address the complex spatial relationships among wind turbines, Hou et al. [36] incorporated a graph neural network (GNN) into the Transformer framework, where each turbine was represented as a node and edges denoted spatial relationships. The results showed that, compared with the Transformer without GNN, the improved model reduced e_{MAE} and e_{RMSE} by 5.85% and 8.95%, respectively. Although these wind power and wind speed forecasting methods perform well in feature extraction, they are limited in handling multimodal data and exhibit weak generalization ability. LLM-based methods for wind power and wind speed forecasting can leverage the reasoning and knowledge transfer capabilities of LLMs to further improve forecasting accuracy.

2.2 Modality alignment

Multimodal data fusion and alignment have become one of the focal areas in current AI research. In computer vision and natural language processing, extensive research has focused on mapping heterogeneous modality features into a unified representation space to enable tasks such as cross-modal retrieval, generation, and reasoning. For example, image–text alignment models can associate image content with textual descriptions, while speech–text alignment methods align audio signals with corresponding transcripts. These alignment techniques jointly learn heterogeneous data distributions, allowing models to establish semantic correspondences across modalities and significantly broadening their application scope [37]. Chen [38] proposed a Model Reprogramming framework that enables pretrained models to be applied to cross-domain tasks through “noise injection + input mapping,” without modifying model parameters. Yang et al. [39] proposed reprogramming acoustic models for time series classification. Their method transforms time series into a format interpretable by acoustic models, thereby achieving modality alignment, and replaces

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training from scratch with "input format editing + model reprogramming."

Unlike image or speech signal processing, wind power data are inherently sparse, making it difficult to establish accurate semantic mappings with textual information [26]. Moreover, LLMs are primarily designed for textual modalities, and the distribution of wind power data differs substantially from that of the text corpora used in LLM pretraining, making it difficult for LLMs to directly process continuous wind power sequences. The wind power modality alignment approach offers a new pathway to bridge wind time series data with the semantic space of LLMs, enabling LLM-based forecasting without the need for extensive fine-tuning on wind power data.

2.3 Large language models

LLMs have demonstrated remarkable generalization capabilities in natural language understanding and generation tasks. Pretrained on massive corpora, these models are capable of memorizing complex sequential patterns and performing previously unseen tasks when given appropriate prompts—exhibiting strong structural generalization and contextual reasoning abilities. Representative LLMs currently used in the time series forecasting domain include Bidirectional Encoder Representations from Transformers (BERT) [40], GPT [41], and Large Language Model Meta AI (LLaMA) [42].

BERT, developed by Google, adopts the encoder structure of the Transformer architecture and focuses on bidirectional context modeling. It learns semantic relationships between tokens through masked language modeling and next-sentence forecasting, making it particularly effective for language understanding tasks. GPT, proposed by OpenAI, also leverages the Transformer architecture but utilizes a decoder-only structure. It generates text in a left-to-right manner by predicting one token at a time based solely on left-side context, making it especially well-suited for text generation and logical reasoning tasks. LLaMA, developed by Meta, is a lightweight LLM designed for high efficiency. It is pretrained on a large-scale multilingual corpus, giving it strong capabilities in cross-lingual understanding and reasoning. Due to differences in training data types, corpus scales, and architectural designs, LLMs vary in their reasoning abilities, generalization performance, robustness, and parameter sizes. Therefore, when applying LLMs to wind power forecasting, the choice of model should be made carefully based on available computational resources, dataset size, and the specific objectives of the forecasting task. In this study, from the perspective of practical wind farm applications, we comprehensively considered computational resources, task complexity, deployability, cost, and data privacy. Accordingly, the GPT2-XL version of the pretrained model was selected, as it provides strong reasoning capability, high compatibility, and a moderate parameter scale suitable for real-world deployment.

3 Methodology

3.1 Modality alignment-driven large language model (MALLM)

The proposed modality alignment-driven large language model (MALLM) for wind power forecasting primarily consists of two core components: a modality alignment module and a pretrained LLM. The main function of the modality alignment module is to transform continuous wind power data into a modality compatible with LLMs while embedding temporal information during the conversion process, thereby avoiding the loss of temporal features. The embedded temporal information primarily includes the acquisition time corresponding to the current power data and the wind speed associated with that moment. The modality alignment module not only enables the LLM to accurately interpret and reason about wind power data but also allows MALLM to directly leverage open-source LLMs for forecasting without retraining or fine-tuning. After obtaining the forecasting results in textual modality, the modality alignment module is executed in reverse to generate continuous numerical forecasting results. The pretrained LLM is primarily responsible for reasoning over the modality-aligned wind power data with embedded temporal information to produce forecasting results in textual modality. The forecasting workflow of MALLM is illustrated in Fig. 1.

As shown in Fig. 1, the forecasting workflow of MALLM consists of five main steps.

Step 1: Raw wind power operation data are collected from the wind farm through the SCADA system. Since the original data contain missing values and outliers, the interquartile range (IQR) method and linear interpolation are applied for imputation and cleaning to construct the final dataset.

Step 2: The modality alignment module maps the power data and their corresponding temporal information into discrete token sequences. After aligning the numerical and textual modalities, the aligned data are input into the LLM for forecasting.

Step 3: The token sequence is fed into the pretrained LLM for inference, producing a sequence of forecast tokens. As the LLM output is in textual modality, the numerical results cannot be directly obtained.

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Step 4: The forecast tokens are processed by the modality alignment module in reverse to recover the corresponding temporal and power values.

Step 5: Through reverse modality alignment, the predicted token sequence is mapped back into numerical form, yielding the final continuous wind power forecasting results.

This workflow enables cross-modality forecasting without retraining or fine-tuning the LLM, thereby effectively reducing the training data requirements and deployment costs of LLM-based wind power forecasting models.

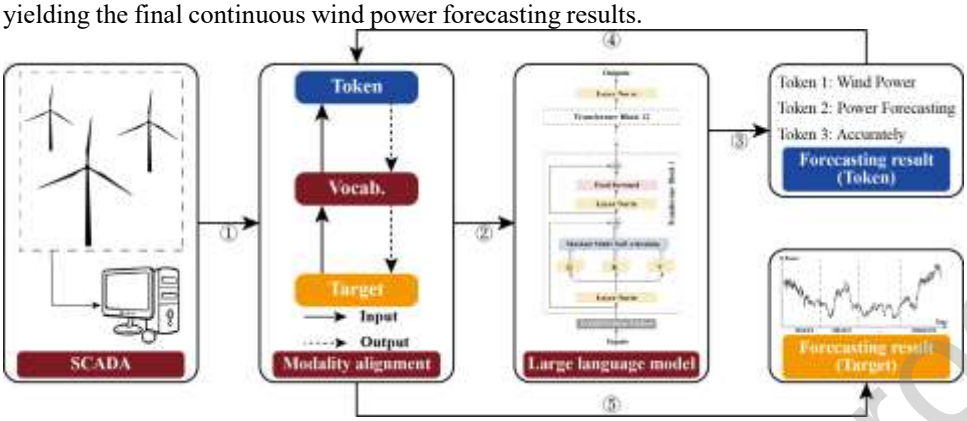

Fig. 1 Workflow of MALLM

3.2 Wind power data modality alignment method

SCADA data collected from wind farms are typically continuous time-series data, which contain both inherent data characteristics and abundant temporal features, representing a typical continuous data modality. LLMs are generally trained on discrete text corpora, making them incapable of directly handling continuous power data. In other LLM application domains, such as speech processing and image recognition, accurate mappings can be established between different modalities—for example, converting audio into text or images into numerical matrices. However, in the context of wind power forecasting, continuous power data cannot be accurately mapped to discrete textual modalities. The proposed principle of data modality alignment is illustrated in Fig. 2. In natural language models, a token is the smallest unit of input that the model can recognize and process. In this paper, a token refers to the minimal semantic unit derived from transforming continuous SCADA time-series data into a textual modality. Each token encodes both feature semantics and temporal information, enabling the large language model to interpret and process time-series data in a linguistically consistent manner.

First, the power data are divided into several segments, each containing a fixed number of power values. The combination of consecutive values captures both magnitude information and short-term variation trends. After modality alignment, each segment is represented by a set of words, which are concatenated into tokens, thereby aligning continuous data with discrete textual modalities. Since the input and output of LLMs are textual, the predicted tokens in discrete form must be reverse-mapped into words, which are then converted into corresponding power values to obtain the final numerical forecasting results.

The framework of data modality alignment is depicted in Fig. 3. Initially, the input power series of length L is partitioned in the data modality alignment module. The sequence is then divided sequentially into N non-overlapping segments $\{s_1, s_2, \ldots, s_N\}$, each of length S. This operation converts the long input sequence into shorter tokens. Each token comprises S consecutive power values. In this way, local trends and variation patterns of the power sequence are retained. This design mitigates the inefficiency of directly modeling long sequences. The entire procedure is formally expressed in Eq. (1).

$$s_i = \{x_{(i-1)S+1}, \ldots, x_{iS}\}, \quad i = 1, \ldots, N, \tag{1}$$

where s_i indicates the k-th segment, while x_i corresponds to the original power series.

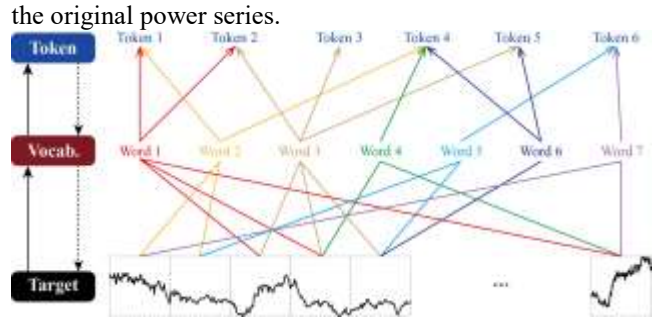

Fig. 2 Schematic diagram of modality alignment

For each s_i, a modality mapping strategy is employed to project it into the vector space of the LLM, as formulated in Eq. (2).

$$\boldsymbol{SE}_i = \text{SegmentEmbedding}(s_i), \quad \boldsymbol{SE}_i \in \mathbb{R}^D, \tag{2}$$

where SegmentEmbedding denotes a function that

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transforms power sequences into their corresponding vector representations. $\boldsymbol{SE}_i$ refers to the vectorized representation of a segment, while D specifies the dimensionality adopted by the LLM.

Wind power generation demonstrates pronounced periodic characteristics. During the segmentation of power data, temporal information is concurrently embedded into each segment. Specifically, the start and end times of segment s_i are transformed into a descriptive token. The resulting token is passed into the LLM encoder, where its positional encoding is extracted and utilized as the embedded temporal representation. The procedure is formally defined in Eq. (3).

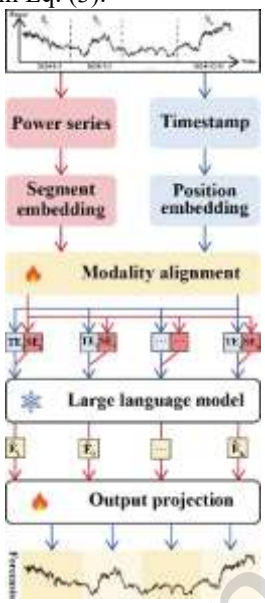

Fig. 3 Diagram of modality alignment

$$\boldsymbol{TE}_i = \mathrm{LLM}(\mathrm{TimestampTemplate}(s_i))_{\mathrm{last}}, \quad \boldsymbol{TE}_i \in \mathbb{R}^D \quad (3)$$

where TimestampTemplate(s_i) denotes the conversion of the time interval of segment s_i into a descriptive textual form $\mathrm{LLM}(\cdot)_{\mathrm{last}}$ corresponds to the positional encoding vector derived from this descriptive text.

Once both the numeric content embedding and the temporal position embedding are obtained, they are integrated to form a discrete representation $\boldsymbol{E}_i$ that is compatible with the LLM. This representation encapsulates both numeric and temporal information, thereby associating numerical values with their corresponding temporal context. The representation $\boldsymbol{E}_i$ is formally defined in Eq. (4).

$$\boldsymbol{E}_i = \boldsymbol{SE}_i + \boldsymbol{TE}_i. \quad (4)$$

The vector $\boldsymbol{E}_i$ is fed into the LLM for forecasting. This forecasting step is described in Eq. (5).

$$\{\boldsymbol{E}_2, \ldots, \boldsymbol{E}_{N+1}\} = \mathrm{LLMLayers}(\boldsymbol{E}_1, \ldots, \boldsymbol{E}_N), \quad (5)$$

where $\boldsymbol{E}_i$ denotes the forecasted token for $\boldsymbol{E}_i$.

During the forecast projection phase, the output $\boldsymbol{E}_i$ generated by the LLM is transformed back into the numerical power domain, yielding the predicted values for future power generation. This mapping procedure is mathematically defined in Eq. (6).

$$\hat{s}_i = \mathrm{SegmentProjection}(\boldsymbol{E}_i), \quad i = 2, \ldots, N+1, \quad (6)$$

where $\hat{s}_i$ denotes the predicted fragment corresponding to time segment s_i.

3.3 Pre-trained large language model

In MALLM, the large language model adopted is GPT2. GPT2 adopts a decoder-only Transformer architecture and is pretrained in an autoregressive manner (from left to right), endowing it with strong contextual modeling and multi-step sequence generation capabilities. In the context of wind power forecasting, GPT2 effectively captures long-term dependencies by leveraging long historical sequences and supports flexible forecasting over arbitrary horizons by predicting future time steps progressively. GPT2 possesses strong contextual modeling and reasoning capabilities, enabling adaptation to various downstream tasks without additional fine-tuning. The architecture of GPT2 is illustrated in Fig. 4.

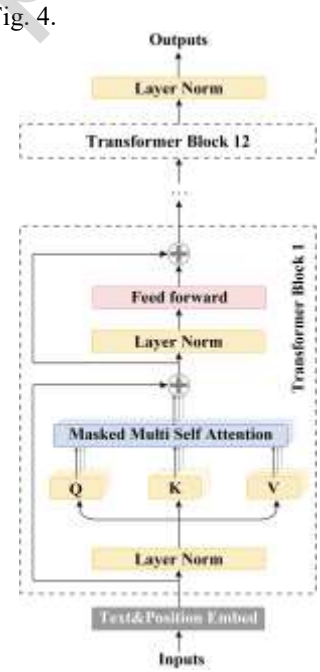

Fig. 4 Diagram of the GPT2 model structure

The input to GPT2 first undergoes token and positional embedding to provide semantic and temporal information for the sequence. The input is then sequentially processed through 12 stacked Transformer decoder blocks, each consisting of masked multi-head self-attention, residual connections, layer normalization, and a feedforward neural network [43]. In the attention mechanism, the input sequence is mapped into query, key, and value vectors, with a masking strategy ensuring that only historical information is used during forecasting, thereby preserving the autoregressive

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property. The deep semantic representations produced by multiple stacked layers are finally passed through the output layer to generate the predicted tokens. When applied within MALLM for wind power forecasting, the self-attention mechanism enables GPT2 to simultaneously capture local and global features. This allows effective modeling of periodicity and trends in modality-aligned wind power data, thereby significantly enhancing forecasting accuracy and stability.

4 Experiments

4.1 Datasets

To validate the effectiveness of MALLM, two real-world SCADA datasets from wind farms located in geographically distinct regions of China were used. The two wind farms are situated in northwestern and southeastern China and are referred to as Dataset 1 and Dataset 2, respectively, according to their order of appearance. The detailed attributes of the datasets are summarized in Table 1. Dataset 1 corresponds to a wind farm with an installed capacity of 46 MW. The data were collected over the entire year of 2023, with a sampling interval of 15 minutes. During the observation period, the recorded power output ranged from 0 to 46 MW, with an average daily power generation of approximately 10.9 MW. Dataset 2 shares the same sampling interval and time span as Dataset 1, but comes from a different wind farm with an installed capacity of 50 MW. The power output in this dataset also ranges from 0 to 50 MW, and the average daily generation is 9.5 MW. The normalized annual power generation curves for both datasets are illustrated in Fig. 5, highlighting the variability and distinct generation patterns under different geographic and climatic conditions.

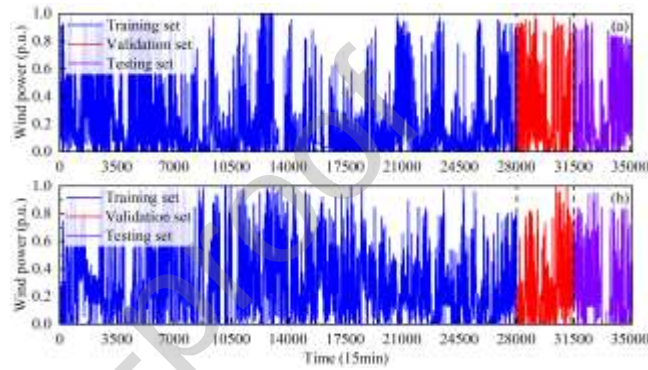

Fig. 5 The power curves of the three datasets: (a) Dataset l, (b) Dataset 2

Table 1 Datasets properties

Datasets	Features	Acquisition time	Sample rate	Sample size	Installed capacity	Average
1	2	2023	15 min	35040	46 MW	10.9 MW
2	2	2023	15 min	35040	50 MW	9.5 MW

As shown in Fig. 5, the normalized power output curves from both datasets used in this study exhibit significant volatility and non-stationarity. These characteristics reflect the influence of wind speed randomness and seasonal variation on wind power generation. To facilitate model training and performance evaluation, each dataset was divided into training, validation, and test sets in a ratio of 8:1:1. The training set was used for parameter learning, the validation set for model selection and hyperparameter tuning, and the test set for final performance assessment. From the figure, it can be observed that Dataset 1 and Dataset 2 share certain seasonal trends in their power output patterns, while also exhibiting different degrees of stochastic variation. Dataset 1 presents sharper short-term fluctuations during specific time periods, whereas Dataset 2 shows smoother and more continuous changes in power output. These differences provide a strong basis for evaluating the robustness of the proposed method under varying wind conditions and geographical settings. This observation is further supported by the monthly statistics of normalized power generation illustrated in Fig. 6.

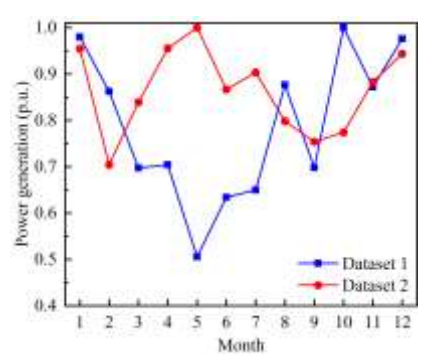

Fig. 6 Monthly power generation curves for the two datasets

For Dataset 1, power generation clearly declines from late spring to summer, reaching a minimum in May, while significantly increasing in autumn and winter—revealing strong seasonal peaks and troughs. In contrast, Dataset 2 exhibits more stable power generation throughout the year, with smaller overall fluctuations, though relatively higher output is still observed during both summer and winter months. These distinctions highlight the geographical, climatic, and seasonal differences between the two wind farms, further demonstrating the appropriateness and representativeness of the selected datasets for evaluating the

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proposed forecasting approach.

4.2 Experiment settings

For the comparative and ablation experiments conducted in this study, two widely adopted evaluation metrics in practical wind farm operations were selected to assess the performance of wind power forecasting models: Mean Absolute Error (MAE) [44] and Root Mean Squared Error (RMSE) [45]. e_{MAE} measures the average of the absolute differences between predicted values and actual values. It provides an intuitive indication of the overall forecasting accuracy—lower e_{MAE} values indicate that the model's forecasts deviate less from the true observations. e_{RMSE}, on the other hand, places greater emphasis on the variability of forecasting errors. Due to the squaring of residuals, it is more sensitive to large deviations and therefore serves as an effective indicator of the model's robustness under extreme conditions or abrupt fluctuations in wind power output. By combining both e_{MAE} and e_{RMSE}, the proposed method can be comprehensively evaluated from multiple perspectives, accounting for both general accuracy and stability under uncertainty. The formulations for e_{MAE} and e_{RMSE} are presented in Eqs. (7) and (8), respectively.

$$e_{\mathrm{MAE}} = \frac{1}{n}\sum_{i=1}^{n}\left|x_i - \hat{x}_i\right|, \tag{7}$$

$$e_{\mathrm{RMSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \hat{x}_i\right)^2}, \tag{8}$$

where n represents the number of samples, $\hat{x}_i$ is the forecasting power value.

To evaluate the performance of MALLM, five state-of-the-art time series forecasting models were selected for comparison: FEDformer [46], Informer [47], Autoformer [48], iTransformer [49], and Transformer [50]. Descriptions of these baseline models are summarized in Table 2.

Although all five baselines are based on the Transformer architecture, each model exhibits distinct design goals and optimization strategies. The Transformer serves as the foundational architecture, providing a unified framework upon which subsequent innovations are built. Informer and FEDformer focus primarily on improving computational efficiency and modeling capability for long sequence forecasting. Informer introduces the ProbSparse attention mechanism to reduce computational overhead, while FEDformer enhances long-sequence modeling through frequency domain decomposition. In contrast, Autoformer and iTransformer emphasize architectural innovations. Autoformer introduces trend-seasonal decomposition and an auto-correlation mechanism to improve interpretability, whereas iTransformer reconstructs attention mechanisms at the variable-level token granularity to strengthen the modeling of multivariate dependencies.

Table 2 Description of the comparison models

Model	Description
FEDformer	Introduces frequency domain enhancement modules, integrating seasonal–trend decomposition with Fourier frequency representations into the Transformer architecture, thereby improving the ability for long-sequence forecasting.
Informer	Employs the ProbSparse self-attention mechanism and a self-attention distillation structure, significantly reducing time and memory complexity while supporting one-pass multi-step forecasting, making it effective for ultra-long sequence forecasting.
Autoformer	Incorporates trend–seasonal decomposition as an internal structure and replaces traditional self-attention with an autocorrelation mechanism, thereby improving efficiency and interpretability. It demonstrates superior performance in long-term forecasting tasks.
iTransformer	Treats each variable in the sequence as an independent token (in contrast to the original Transformer), restructuring the Transformer to enhance the model's capability in capturing inter-variable dependencies, thus achieving stronger generalization and high-dimensional forecasting performance.
Transformer	A classical sequence model based on multi-head self-attention and positional embeddings, with strong parallel processing capability, serving as the foundational architecture for numerous Transformer variants.

In the experiments, each model was trained and tested according to the predefined training, validation, and test set partitions. As all models adopt an encoder–decoder architecture, the number of encoder and decoder layers was uniformly set to three, with a learning rate of 0.0001. Other model-specific hyperparameters are listed in Table 3. An early stopping strategy was employed during training to prevent overfitting and improve training efficiency.

Table 3 Hyper-parameter of the comparison models

Model	Hyper-parameters
MALLM	Hidden dimension of LLM: 768; number of LLM hidden layers:64
FEDformer	Head number of attention: 16; number of modes for the Fourier block: 64
Informer	Head number of attention: 16; number of windows to sample in each inference batch: 1024
Autoformer	Head number of attention: 16
iTransformer	Head number of attention: 16
Transformer	Head number of attention: 16

4.3 Comparative experiment

The comparative performance of MALLM and the baseline models is presented in Tables 4 and 5, which report the e_{MAE} and e_{RMSE} results, respectively. Overall, MALLM

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consistently achieves the lowest error metrics across both datasets, significantly outperforming the other models. As shown in Table 4, the average e_{MAE} of MALLM on Dataset 1 is 1.0089, notably lower than that of the second-best model, Informer, which records an e_{MAE} of 1.1410—representing a maximum reduction of 15.57% compared to other models. On Dataset 2, MALLM achieves an average e_{MAE} of 1.1578, again outperforming all baseline models, with improvements ranging from a minimum reduction of 11.27% to a maximum of 15.50%. Table 5 reveals a similar trend for e_{RMSE}. On Dataset 1, MALLM achieves an average e_{RMSE} of 1.7663, which is 15.94% lower than that of the worst-performing baseline, Transformer. For Dataset 2, the e_{RMSE} values of the comparison models, ranked from lowest to highest, are: Autoformer, Transformer, FEDformer, Informer, and iTransformer. MALLM yields a 9.94% lower e_{RMSE} than Autoformer, and a 15.32% reduction compared to iTransformer. Taken together, these results indicate that MALLM achieves the best overall forecasting accuracy across both datasets. It consistently obtains the lowest e_{MAE} and e_{RMSE}, demonstrating strong generalization capability and high forecasting precision in different wind farm settings. Regarding performance across different forecasting horizons, error metrics for all models increase with longer forecasting steps. However, MALLM exhibits a more gradual increase in both e_{MAE} and e_{RMSE}, indicating stronger stability and error control. For instance, on Dataset 1, when increasing the forecast horizon from 1 to 24 steps, MALLM's e_{MAE} and e_{RMSE} increase by 54.14% and 54.56%, respectively, whereas the largest observed increases in the baseline models are 60.27% (e_{MAE}) and 55.43% (e_{RMSE}). This suggests that MALLM accumulates less error over longer horizons, reinforcing its robustness in multi-step forecasting scenarios.

Table 4 e_{MAE} of different models on two datasets

Dataset	Time step	MALLM	FEDformer	Informer	Autoformer	iTransformer	Transformer
1	1	0.6076	0.6758	0.7374	0.6429	0.7047	0.7414
	4	0.7404	0.8256	0.9049	0.8649	0.8799	0.8243
	8	0.8709	1.0415	0.9157	1.0325	1.0280	0.9440
	12	1.0197	1.2007	1.1748	1.0741	1.1164	1.1842
	16	1.2135	1.4895	1.3755	1.3985	1.4470	1.5014
	20	1.2379	1.5167	1.4048	1.4459	1.4585	1.5721
	24	1.3723	1.4737	1.4738	1.6180	1.5302	1.5974
	Average	1.0089	1.1748	1.1410	1.1538	1.1664	1.1950
2	1	0.6886	0.7265	0.8556	0.8504	0.7596	0.8190
	4	0.8230	0.8819	0.9056	1.0086	0.9043	0.9798
	8	0.9740	1.0288	1.1196	1.0729	1.2122	1.0927
	12	1.2392	1.4589	1.3757	1.4647	1.3986	1.3739
	16	1.3941	1.5515	1.5432	1.6917	1.7125	1.6895
	20	1.4366	1.6546	1.5190	1.6680	1.6898	1.7412
	24	1.5490	1.9076	1.8153	1.7905	1.8727	1.8951
	Average	1.1578	1.3157	1.3049	1.3638	1.3642	1.3702

Table 5 e_{RMSE} of different models on two datasets

Dataset	Time step	MALLM	FEDformer	Informer	Autoformer	iTransformer	Transformer
1	1	1.0911	1.2001	1.2553	1.1984	1.2553	1.3449
	4	1.3754	1.5571	1.4583	1.4698	1.6029	1.5848
	8	1.5577	1.8710	1.7224	1.9151	1.7890	1.7918
	12	1.8293	2.0045	2.2531	2.2502	1.9922	2.2128
	16	2.0216	2.1538	2.2195	2.3787	2.4148	2.3855
	20	2.0880	2.3134	2.2529	2.3340	2.3097	2.4855
	24	2.4012	2.5987	2.7563	2.6890	2.5329	2.9034
	Average	1.7663	1.9569	1.9883	2.0336	1.9853	2.1012
2	1	1.2782	1.4800	1.5038	1.5718	1.5794	1.4808
	4	1.4329	1.7359	1.6861	1.6783	1.6273	1.5867
	8	1.7094	2.1012	1.9780	1.7980	2.1253	1.9969
	12	2.0161	2.2451	2.1533	2.1578	2.5055	2.1292
	16	2.2928	2.4579	2.7905	2.7117	2.7986	2.4246
	20	2.4907	2.7288	2.7750	2.6178	2.7619	3.0250
	24	2.7321	3.1021	2.9706	2.9566	3.0791	3.0655
	Average	1.9932	2.2644	2.2653	2.2131	2.3539	2.2441

In addition to evaluating e_{MAE} and e_{RMSE} results, the effectiveness of MALLM can also be intuitively assessed from the overall forecasting trends and error magnitudes. Taking one day from the test set as an example, the forecasting results and corresponding error curves are illustrated in Figs. 7 and 8. As shown in Fig. 7, MALLM consistently produces accurate forecasts across different datasets and forecasting horizons. When the forecasting

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horizon is set to one step, all models capture the real power variations reasonably well, but MALLM yields the smallest deviations from the ground truth. As the forecasting horizon extends to 24 steps, deviations and fluctuations increase for all models; however, MALLM exhibits relatively smaller deviations, indicating stronger stability and generalization in long-term forecasting.

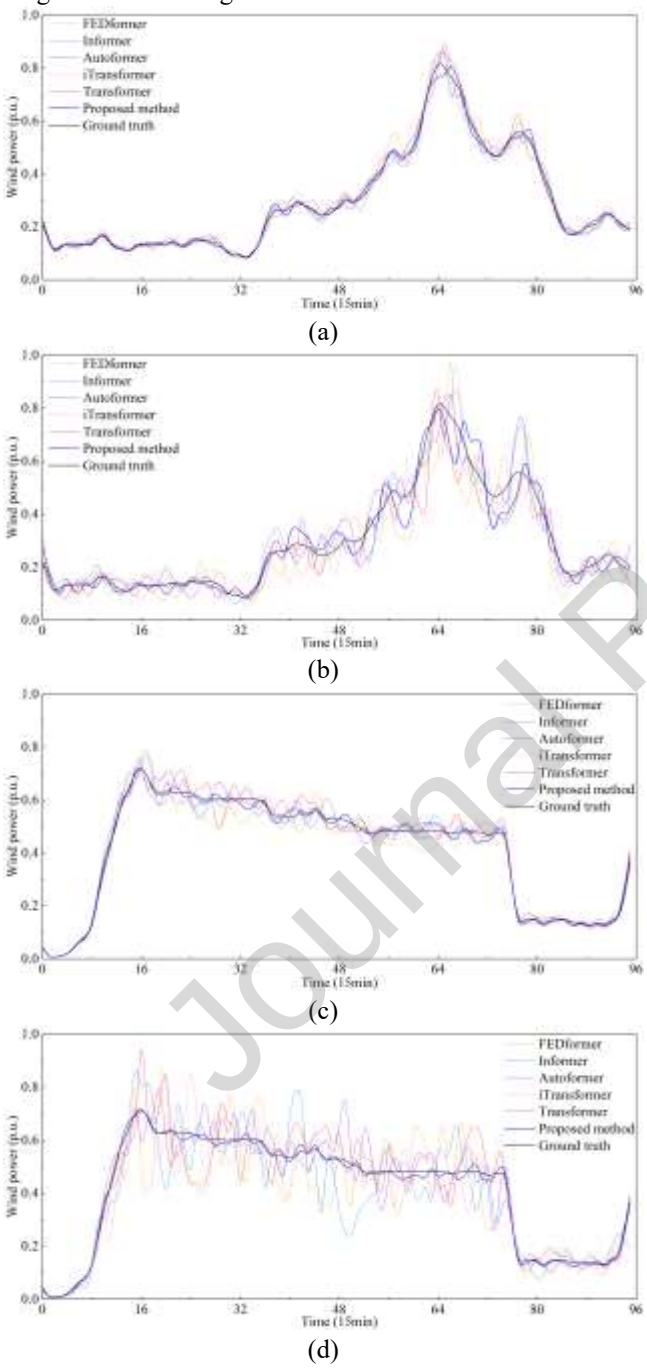

Fig. 7 Forecasting results and the ground truth of different datasets: (a) Dataset 1, 1 time step, (b) Dataset 1, 24-time steps, (c) Dataset 2, 1 time step, (d) Dataset 2, 24-time steps

Fig. 8 further substantiates this conclusion. At the one-step horizon, the errors of all models fluctuate around zero, but MALLM produces the smallest error magnitude with the smoothest curve, whereas the other models show more pronounced alternating positive and negative deviations. As the forecasting horizon increases, the error curves of the baseline models show large oscillations, while MALLM consistently maintains a relatively narrow error range. These results demonstrate that the proposed wind power data modality alignment method can effectively transform continuous power data into discrete textual modalities, enabling accurate and efficient forecasting with LLMs.

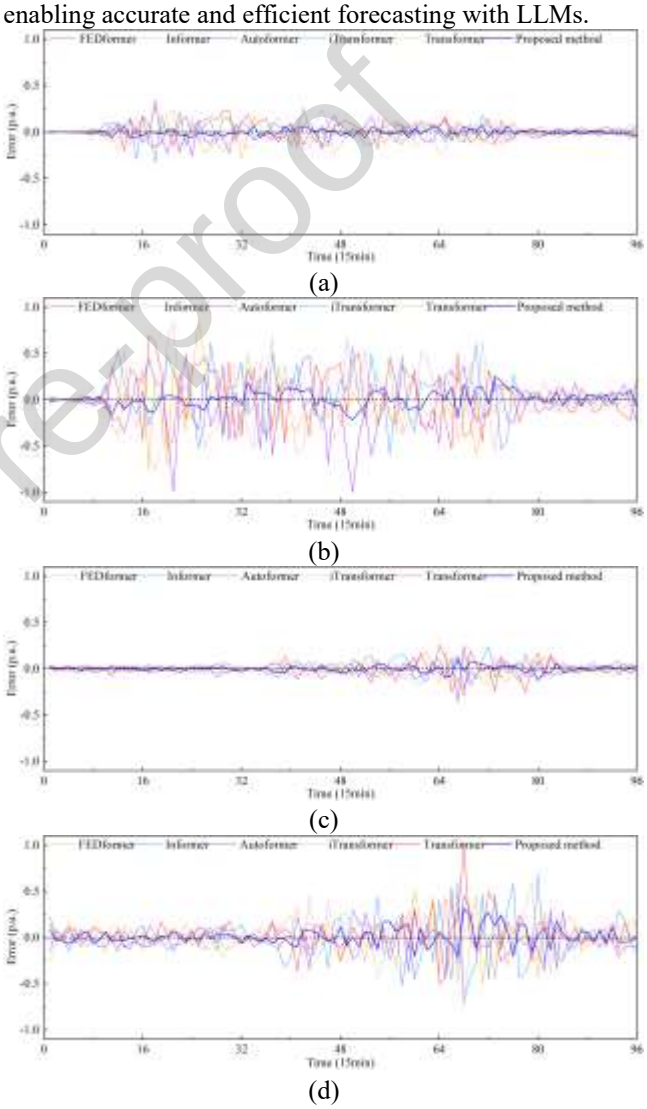

Fig. 8 Error of forecasting results for the two datasets: (a) Dataset 1, 1 time step, (b) Dataset 1, 24-time steps, (c) Dataset 2, 1 time step, (d) Dataset 2, 24-time steps

4.4 Ablation studies

To assess the impact of LLM parameter scale on the forecasting accuracy of MALLM, we conducted ablation experiments on both datasets using different versions of the

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GPT2 model with varying parameter sizes. Detailed specifications of the selected models are listed in Table 6. Three GPT2 variants with the same architecture but different parameter scales were selected for comparison. These models are referred to as GPT2-XL, GPT2-Large, and GPT2-Medium, corresponding to parameter sizes of 1.5 billion, 0.8 billion, and 355 million, respectively.

Table 6 Detailed Properties of GPT2

Name	Version	Parameter scale	Release time
GPT2-XL	XL	1.5 B	2019
GPT2-Large	Large	0.8 B	2019
GPT2-Medium	Medium	355 M	2019

The ablation results regarding the parameter scale of LLMs are presented in Tables 7 and 8. In terms of error trends, regardless of the parameter scale of GPT2, forecasting errors gradually increase as the forecasting horizon extends. In terms of error magnitude, forecasting errors decrease progressively with the increase of GPT2 parameter scale. On Dataset 1, GPT2-XL achieved an average e_{MAE} of 1.0089, representing reductions of 1.48% and 2.06% compared with GPT2-Large and GPT2-Medium, respectively. Its average e_{RMSE} was 1.7663, showing reductions of 2.01% and 2.45% compared with GPT2-Large and GPT2-Medium. Similar results were observed on Dataset 2: the average e_{MAE} of GPT2-XL was reduced by 1.65% and 2.28% compared with GPT2-Large and GPT2-Medium, respectively, while its average e_{RMSE} was reduced by 2.14% and 2.58%. These results demonstrate that the proposed MALLM is compatible with LLMs of different parameter scales, though enlarging the model scale provides only limited improvements in forecasting accuracy. This finding further indicates that the proposed modality alignment method effectively bridges continuous power data with discrete textual modalities, thereby reducing MALLM's dependence on large-scale LLMs and enhancing its computational efficiency. It also suggests that pretrained LLMs lack domain-specific knowledge of wind power forecasting, which constrains their reasoning capability over modality-aligned wind power data. If sufficient data and domain knowledge are available to meet fine-tuning requirements, LLMs should be fine-tuned to further enhance forecasting accuracy.

In the MALLM, the selection of a suitable LLM is primarily influenced by two factors: model architecture and parameter scale. The model architecture determines whether the LLM is structurally appropriate for wind power forecasting—particularly its capability for sequence generation and modeling long-term dependencies. The parameter scale, on the other hand, affects the model's deployability under real-world operational conditions. In practice, the choice of LLM should be guided by a comprehensive evaluation of the available computational resources and the quality of SCADA data in the wind farm, to ensure an optimal balance between forecasting accuracy and deployment feasibility.

Table 7 e_{MAE} of ablation studies on two datasets

Dataset	Models	15 min	1 h	2 h	3 h	4 h	5 h	6 h	Average
1	GPT2-XL	0.6076	0.7404	0.8709	1.0197	1.2135	1.2379	1.3723	1.0089
	GPT2-Large	0.6157	0.7536	0.8821	1.0314	1.2334	1.2621	1.3901	1.0241
	GPT2-Medium	0.6191	0.7565	0.8918	1.0415	1.2383	1.2671	1.3967	1.0301
2	GPT2-XL	0.6886	0.8230	0.9740	1.2392	1.3941	1.4366	1.5490	1.1578
	GPT2-Large	0.7004	0.8405	0.9938	1.2535	1.4149	1.4621	1.5754	1.1772
	GPT2-Medium	0.7054	0.8438	0.9994	1.2651	1.4251	1.4699	1.5852	1.1848

Table 8 e_{RMSE} of ablation studies on two datasets

Dataset	Models	15 min	1 h	2 h	3 h	4 h	5 h	6 h	Average
1	GPT2-XL	1.0911	1.3754	1.5577	1.8293	2.0216	2.0880	2.4012	1.7663
	GPT2-Large	1.1131	1.4064	1.5903	1.8548	2.0568	2.1397	2.4571	1.8026
	GPT2-Medium	1.1155	1.4151	1.5966	1.8758	2.0786	2.1421	2.4515	1.8107
2	GPT2-XL	1.2782	1.4329	1.7094	2.0161	2.2928	2.4907	2.7321	1.9932
	GPT2-Large	1.3087	1.4648	1.7551	2.0614	2.3334	2.5524	2.7815	2.0368
	GPT2-Medium	1.3142	1.4705	1.7607	2.0635	2.3478	2.5596	2.8056	2.0460

4.5 Challenges and limitations

Currently, data-driven wind power forecasting methods have achieved significant improvements in accuracy, and many enhanced deep learning models have shown strong adaptability to real-world wind farm scenarios. Although several studies have explored the use of LLMs in wind power forecasting with promising results, there remain key challenges and limitations to be addressed in practical applications:

In terms of data quality and availability, real-world SCADA systems often suffer from missing values, noise, or sensor malfunctions, all of which can degrade model performance and limit generalizability.

Regarding computational cost, training or fine-tuning

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large LLMs requires substantial computational resources, which can hinder deployment in environments that demand real-time forecasting.

With respect to data diversity, most existing models rely on a single data source, such as SCADA or meteorological data alone. This limits the model's ability to fully capture the complex meteorological factors that influence wind power generation. Taking MALLM as an example, while it demonstrates superior forecasting accuracy compared to all baseline models, its computational cost is relatively high. The operation information of different models is shown in Table 9.

Table 9 Operation information of different models

Model	Time (s/epoch)	e_{MAE}	e_{RMSE}
MALLM	54.9	1.0089	1.7663
FEDformer	18.3	1.1748	1.9569
Informer	27.4	1.1410	1.9883
Autoformer	31.6	1.1538	2.0336
iTransformer	14.7	1.1664	1.9853
Transformer	38.1	1.1950	2.1012

As shown in Table 9, on Dataset 1, MALLM's runtime is approximately 44.09% longer than that of the Transformer model (the slowest among the baselines), and about 2.7 times longer than iTransformer, which is the fastest. Therefore, when deploying forecasting models in operational wind farms, it is essential to strike a balance between accuracy and computational efficiency, selecting models that best align with available resources and deployment requirements.

5 Conclusions

(1) A modality alignment–driven large language model for wind power forecasting was developed without fine-tuning or retraining the pretrained LLM. Experimental results demonstrate that the proposed framework can effectively utilize the reasoning capability of LLMs for wind power prediction under limited data conditions.

(2) The modality alignment strategy, combined with temporal information embedding, enables the transformed power data to retain essential seasonal and periodic patterns. Comparative experiments show that this design significantly improves forecasting accuracy compared with existing state-of-the-art models.

(3) On Dataset 1, MALLM achieves an average e_{MAE} of 1.0089, corresponding to a reduction of approximately 15.6% relative to the best-performing baseline model. On Dataset 2, the average e_{MAE} is reduced to 1.1578, yielding an error reduction of up to 15.5%. In addition, MALLM consistently produces the lowest e_{RMSE} across both datasets, indicating superior prediction stability.

(4) Ablation studies reveal that increasing the parameter scale of the underlying pretrained LLM leads to only marginal performance gains. This result indicates that the forecasting performance of MALLM is not strongly dependent on the size of the base model, highlighting its adaptability and suitability for practical deployment scenarios with constrained computational resources.

(5) The experimental findings collectively demonstrate that MALLM provides a robust and efficient solution for wind power forecasting by alleviating the dependence on large-scale training data and avoiding the high computational cost associated with LLM fine-tuning.

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CRediT authorship contribution statement

Ling Xiang and Yue Zhang were in charge of the overall research design and supervised the whole study. Yang Chen developed the methodology and wrote the original draft of the manuscript. Weiqing Jia performed data preprocessing and software implementation. Ruibin Ban conducted the formal analysis and validation of the proposed method. Guopeng Zhu assisted with data acquisition, sampling, and laboratory analyses. Aijun Hu reviewed and edited the manuscript and provided critical revisions. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare no competing financial interests.

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